

VISHAL GUPTA

Mumbai 400068 | +91 7977221647 | vishal210496@gmail.com | linkedin.com/in/vishal-gupta-818b64114 | vishal-gupta359.github.io

PROFESSIONAL SUMMARY

Results-driven Backend Developer with 7+ years of IT experience, including 4+ years specializing in Python (Django/Flask) and RASA-based conversational AI. Expertise in building scalable chatbot platforms and real-time customer engagement systems. Promoted to Senior Software Developer for consistently delivering high-performance, enterprise-grade solutions. Proficient in AWS cloud deployments, Generative AI solutions using AWS Bedrock (RAG, Knowledge Bases, Agents), and asynchronous messaging with RabbitMQ and Redis, with strong cross-functional collaboration skills.

PROFESSIONAL EXPERIENCE

Senior Software Developer | Route Mobile Ltd, Mumbai

April 2024 – Present

RouBot Platform – Chatbot Builder & Automation

- Contributed to the development of RouBot, a no-code/low-code chatbot builder platform enabling businesses to create and deploy chatbots across WhatsApp, web, and other messaging channels.
- Developed and maintained backend services using Django REST Framework to power chatbot flow configuration, webhook management, and third-party integration APIs.
- Implemented Redis caching strategies to reduce API response times and improve platform scalability under high-traffic conditions.
- Integrated RabbitMQ as a messaging queue to handle asynchronous task processing and decouple microservices, improving system reliability and throughput.
- Collaborated with the AWS team on infrastructure setup and deployments; gained hands-on experience with AWS services including EC2 for compute provisioning and CloudWatch for application monitoring, logging, and alerting.
- Participated in code reviews and enforced coding best practices, providing technical guidance to junior developers.
- Identified and resolved performance bottlenecks in existing API endpoints to enhance overall system responsiveness.

AWS Bedrock – Generative AI Integration

- Integrated AWS Bedrock APIs to incorporate large language model (LLM) capabilities into the platform, enabling intelligent, context-aware conversational responses for chatbot workflows.
- Implemented Retrieval-Augmented Generation (RAG) architecture by connecting Bedrock models with a vector database (Amazon OpenSearch Serverless) to enable semantic search and accurate knowledge retrieval.
- Created and configured Amazon Bedrock Knowledge Bases by ingesting structured and unstructured documents from Amazon S3 buckets, enabling the AI agent to answer queries grounded in proprietary business data.
- Built and deployed Bedrock Agents to automate multi-step conversational workflows, linking them with S3-based data sources and Lambda action groups for dynamic API orchestration.
- Worked with foundation models on Bedrock (Claude, Titan) and tuned prompt templates and inference parameters to optimize response quality and latency.

Software Developer | Route Mobile Ltd, Mumbai

September 2022 – March 2024

Conversational AI – Chatbot Development (RASA)

- Developed intelligent chatbots for WhatsApp and web channels using the RASA framework, addressing diverse business use cases including customer support, lead qualification, and FAQ automation.

- Designed and optimized NLU pipelines and dialogue management flows to improve chatbot intent recognition accuracy and overall user experience.
- Introduced novel strategies to enhance customer interface experiences and optimize chatbot performance.
- Proactively identified and resolved technical challenges during development cycles, reducing bot failure rates.

Conversational Live Agent Platform – Multichannel Customer Support

- Worked on developing a comprehensive customer support solution encompassing a multichannel interaction panel that facilitates real-time agent-customer engagement via WhatsApp, web, and other messaging channels for effective query resolution.
- Built backend APIs using Django REST Framework to manage agent sessions, conversation routing, and real-time message delivery across channels.
- Integrated WhatsApp Business API and Google Business Messages (GBM) to enable seamless real-time communication for support agents.
- Utilized Redis for session management and real-time data caching; worked with PostgreSQL for structured conversation and user data storage.
- Contributed to an analytics dashboard providing data-driven insights into agent performance, response times, and customer satisfaction metrics.
- Coordinated with the DevOps team during migration of applications to AWS infrastructure.
- Collaborated with cross-functional teams to meet project milestones and provided mentorship to junior developers.

Software Engineer | Onward Technologies Ltd, Mumbai

September 2018 – August 2022

Adobe Experience Manager (AEM) – Mathworks

- Developed and maintained content management systems and web-based applications for Mathworks using Adobe Experience Manager (AEM 6.4 and 6.5).
- Created and managed responsive web pages using HTML, CSS, and Bootstrap, ensuring consistent rendering across all devices and browsers.
- Utilized AEM Multi-Site Manager (MSM) to develop live copies from blueprints for efficient multi-region content management and localization.
- Conducted quality checks and audits to identify and resolve issues including broken links, functionality errors, and usability problems.
- Managed page lifecycle operations (create, edit, delete, move) and tracked bugs and project tasks using JIRA.

TECHNICAL SKILLS

Languages: Python

Frameworks & Libraries: Django REST Framework, Flask, RASA 2.x

Databases: PostgreSQL, MySQL, Redis

Messaging & Queues: RabbitMQ, Redis Pub/Sub

Cloud & DevOps: AWS (EC2, CloudWatch, S3, Bedrock, OpenSearch Serverless, Lambda), Docker, Git

APIs & Integrations: WhatsApp Business API, Google Business Messages (GBM), REST APIs

CMS & Other Tools: Adobe Experience Manager (AEM 6.4/6.5), JIRA, HTML, CSS, Bootstrap

EDUCATION

Bachelor of Science in Information Technology — July 2014 – May 2017

Mumbai University, Mumbai